

A special kind of polynomial problem

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Let's begin with an example:

Example 1 (MN Math League 1D 2023/4). Suppose $f(x) = ax^2 + bx + c$, and let $g(x) = x + 6$. If $f(2) = 1$, $f(11) = g(11)$, and $f(23) = g(23)$, find $f(29)$.

Your first instinct might be to set up a system of equations:

$$\begin{aligned}f(2) &= 4a + 2b + c = 1 \\f(11) &= 121a + 11b + c = 17 = g(11) \\f(23) &= 529a + 23b + c = 29 = g(23).\end{aligned}$$

But this is disgusting. Even after you compute a , b , and c (without making a mistake!), you have to find $841a + 29b + c$. Nope.

There's a better way. We're going to need two lemmas:

Theorem 2 (Factor theorem). If $x = r$ is a root of a polynomial P , i.e., $P(r) = 0$, then $(x - r)$ is a factor of P .

Theorem 3 (Fundamental theorem of algebra). A polynomial of degree n has n roots.

This basically means that if you know a polynomial's roots, then you know the polynomial, up to some constant factor. For example, if I gave you a quadratic P with roots 1 and 4, you could deduce that $P(x) = a(x - 1)(x - 4)$ for some constant a . Furthermore, if I told you $P(0) = 1$, you could solve for a : $1 = P(0) = a(0 - 1)(0 - 4)$, which implies that $a = \frac{1}{4}$.

Now we can solve our original problem. The trick is to define the polynomial $P(x) := f(x) - g(x)$. The conditions $f(11) = g(11)$ and $f(23) = g(23)$ tell us that $P(11) = f(11) - g(11) = 0$, and $P(23) = f(23) - g(23) = 0$. So P is a quadratic with roots 11 and 23, i.e., $P(x) = a(x - 11)(x - 23)$ for some constant a . Now, we can apply our condition $f(2) = 1$ to compute a . We have

$$1 = f(2) = P(2) + g(2) = a(2 - 11)(2 - 23) + 2 + 6 = 189a + 8.$$

Solving gives $a = -\frac{1}{27}$, which we can now use to find $f(29)$:

$$f(29) = P(29) + g(29) = -\frac{1}{27}(29 - 11)(29 - 23) + 29 + 6 = \boxed{31}.$$

No system of equations required! Now try this.

Problem 4. Suppose P is a polynomial of degree 2022 satisfying $P(1) = 1$, $P(2) = 2$, $P(3) = 3$, $\dots$, $P(2022) = 2022$. Also, $P(2023) = 1$. Compute $P(2024)$.